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TIME-DEPENDENT FORCE-FREE PULSAR MAGNETOSPHERES: AXISYMMETRIC AND OBLIQUE ROTATORS

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ABSTRACT

Magnetospheres of many astrophysical objects can be accurately described by the low-inertia (or "force-free") limit of MHD. We present a new numerical method for the solution of equations of force-free relativistic MHD based on the finite-difference time-domain approach, with a prescription for handling the spontaneous formation of current sheets. We use this method to study the time-dependent evolution of pulsar magnetospheres in both aligned and oblique magnetic geometries. For the aligned rotator, we confirm the general properties of the time-independent solution of Contopoulos et al. For the oblique rotator, we present the three-dimensiional structure of the magnetosphere and compute, for the first time, the spin-down power of pulsars as a function of the inclination of the magnetic axis. We find that the pulsar spin-down luminosity is $L \approx (\mu^2\Omega_*^4/c^3)(1 + \sin^2\alpha)$, where μ is the stellar dipole moment, Ω_* is the rotation frequency, and α is the magnetic inclination angle. We also discuss the effects of current sheet resistivity and reconnection on the structure and evolution of the magnetosphere.

Subject headings: MHD — pulsars: general — stars: magnetic fields — stars: neutron

1. INTRODUCTION

Many astrophysical objects, including neutron stars and accretion disks, form highly magnetized magnetospheres. Modeling of the structure of such magnetospheres requires solving for the self-consistent behavior of plasma in strong fields, where field energy can dominate the energy in the plasma. This is difficult to do with standard numerical methods for MHD that are forced to evolve small plasma inertial terms. In these cases, it is possible to reformulate the problem and, instead of solving for the plasma dynamics in strong fields, solve for the dynamics of fields in the presence of conducting plasma. This is the approach of force-free relativistic MHD or force-free electrodynamics (FFE; see, e.g., Komissarov 2002). In this Letter we present a new numerical method for solving the equations of FFE and apply it to the problem of the structure of pulsar magnetospheres. We calculate the shape of the magnetosphere for arbitrary magnetic inclination and determine the pulsar spin-down power.

2. NUMERICAL METHOD

The equations of FFE can be derived as the low-inertia limit of MHD (Komissarov 2002). When the plasma inertia and pressure are small, the balance of forces on the plasma is $\rho\boldsymbol{E} + (1/c)\boldsymbol{j} \times \boldsymbol{B} = 0$ (the "force-free" condition), where ρ and $\boldsymbol{j}$ are the charge and current densities. When the plasma is conducting perfectly and so abundant as to short out the accelerating electric fields ($\boldsymbol{E} \cdot \boldsymbol{B} = 0$), the Maxwell equations in flat space, together with the force-free condition, give (Gruzinov 1999; Blandford 2002)

$$\frac{1}{c}\frac{\partial \boldsymbol{E}}{\partial t} = \nabla \times \boldsymbol{B} - \frac{4\pi}{c}\boldsymbol{j}, \quad \frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t} = -\nabla \times \boldsymbol{E}, \tag{1}$$

$$\boldsymbol{j} = \frac{c}{4\pi}\nabla \cdot \boldsymbol{E}\frac{\boldsymbol{E} \times \boldsymbol{B}}{B^2} + \frac{c}{4\pi}\frac{(\boldsymbol{B} \cdot \nabla \times \boldsymbol{B} - \boldsymbol{E} \cdot \nabla \times \boldsymbol{E})\boldsymbol{B}}{B^2}. \tag{2}$$

Equation (2) is the prescription for the plasma current that is needed to satisfy the constraints. The two terms in the equation each have a simple meaning: the first term is the current perpendicular to field lines, given by the $\boldsymbol{E} \times \boldsymbol{B}$ advection of the charge density, while the second term is the parallel current, reducing to $(\nabla \times \boldsymbol{B})_\parallel$ in a steady state. The system (eqs. [1] and [2]) is hyperbolic, evolutionary, and is much simpler than the full MHD equations.

In order to numerically solve this system, we utilize its close relation to Maxwell equations and use the finite-difference time-domain (FDTD) method from electrical engineering (Yee 1966). The method's strengths are its very low numerical dissipation and conservation of a divergence-free condition to machine accuracy. Electric and magnetic fields are decentered componentwise on an orthogonal grid, with the electric components defined in the middle of the cell edges and with the magnetic components defined in the middle of the cell faces, allowing us to compute the change of flux through the cell face from the circulation around the cell edges. We use a third-order Runge-Kutta time integration, maintaining the time alignment of the fields. To calculate the current in equation (2), we use a linear interpolation of the fields to the same locations on the grid (at the E-points). The second term in equation (2) comes from the perfect conductivity constraint and requires the interpolation of both fields and field derivatives. Instead of using the full equation (2), we solve an even simpler system: we advance Maxwell equations with only the perpendicular component of the current, and then after every time step we subtract accumulated $E_\parallel$ from the electric field to enforce $\boldsymbol{E} \cdot \boldsymbol{B} = 0$. This is equivalent to replacing the parallel current with a resistive term $\sigma_\parallel E_\parallel$ (Komissarov 2004), where the parallel conductivity $\sigma_\parallel$ is taken to infinity.

There are two conditions for the validity of the force-free approximation; i.e., the electromagnetic energy density vastly exceeds the plasma pressure and inertia, and the plasma drift velocity is subluminal ($E^2 \leq B^2$). The system (eqs. [1] and [2]) does not enforce these conditions, and the initial force-free configuration may develop regions that violate the approximation. When E exceeds B, as may happen during the interaction of strong waves, the Alfvén speed becomes imaginary and leads to an exponential instability, which physically would be averted by dissipation. Another example is the formation of current sheets, which is a generic feature of force-free evolu-

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tion. In current sheets, the magnetic field can reverse and go through zero. This means that the assumption of magnetic dominance is invalid and that plasma pressure must sustain the current sheet. However, in the framework of FFE, pressure is not included, so we try to keep the current sheets unresolved on the grid, with a jump in fields over one cell. The FDTD method can support such sharp discontinuities without spreading them. In order to capture current sheets, we enforce the condition $E \leq B$ after every time step by resetting the electric field to $E = B$ in the regions where $E > B$. The physics behind this drastic measure is related to dissipation processes that will happen if the $\boldsymbol{E} \times \boldsymbol{B}$ drift approximation is violated and particles accelerate while decoupling from magnetic field lines. This dissipation will restore $E = B$ on plasma timescales; however, the microphysics remains to be clarified. We parameterize this with prescriptions for the dissipative current $j_\perp$ that is added to equation (2) when $E > B$. The prescription that we use can be formulated as $j_\perp = \sigma_\perp E$ when $E > B$ and $j_\perp = 0$ otherwise, where the perpendicular conductivity $\sigma_\perp \to \infty$ (cf. Komissarov 2004).

There are other ways to solve the FFE. By formulation, our method is similar to Komissarov's (2004, 2006) in that we also solve the nonconservative equations with explicit resistive current prescriptions. The limiter for enforcing $E < B$ is similar to what was used in the McKinney (2006a) conservative scheme. Our FDTD discretization has an inherently much lower numerical diffusivity and can develop and sustain sharp current sheets, unlike the Komissarov (2006) scheme, without the need to explicitly null the inflow velocity into current sheets as in McKinney (2006a). In fact, the location of the sheet does not have to be known in advance, which allows for the evolution of complicated initial conditions. Below we describe the application of our code to pulsars.

3. ALIGNED PULSAR MAGNETOSPHERE

Particle extraction from the stellar surface due to unipolar induction and subsequent pair creation is likely to populate the pulsar magnetosphere with abundant, nearly force-free plasma (Goldreich & Julian 1969). The detailed structure of the magnetosphere remained a puzzle due to the difficulty of finding an analytic solution. The first numerical solution to the time-independent set of force-free equations (the "pulsar equation") was found by Contopoulos et al. (1999, hereafter CKF) and was later improved on by Gruzinov (2005a). The CKF solution consists of a region of closed field lines extending to the light cylinder (the "closed zone") and an "open zone" with asymptotically monopolar poloidal field lines. The requirement that the closed zone extend to the light cylinder was imposed from the outset and later relaxed by Goodwin et al. (2004), Contopoulos (2005), and Timokhin (2006), who found steady state solutions with closed zones of arbitrary extent within the light cylinder. This raises the question as to which solution would be chosen by a time-dependent evolution. Recently, Komissarov (2006) used a time-dependent force-free code and a relativistic MHD code for this problem. While his force-free code had converged to a solution with all closed field lines, presumably due to excessive numerical diffusion, the MHD code produced a solution remarkably similar to CKF's. The force-free code of McKinney (2006b) also converged to the CKF result. Generally, we are in agreement with these results; however, the intrinsic low-dissipation nature of our code allows us to find features that were missing in prior studies.

We discretize FFE equations in spherical coordinates on a uniform 1400×181 (r, θ)-grid with an absorbing zone to prevent reflections from the edge of the domain. The star has a radius $R_* = 0.2R_{\rm LC}$, and the light cylinder $R_{\rm LC} \equiv c/\Omega_*$ is at 300 cells. The magnetic field is initialized to be a dipole $B_r = 2\mu \cos\theta/r^3$, $B_\theta = \mu \sin\theta/r^3$, and $B_\phi = 0$, where μ is the magnetic moment. The electric field on the star is set to $\boldsymbol{E} = -\boldsymbol{\Omega}_* \times \boldsymbol{r} \times \boldsymbol{B}/c$ to simulate a rotating conducting sphere, and the rotation is turned on as a step function at $t = 0$. During the initial evolution, a torsional Alfvén wave is emitted from the surface (Spitkovsky 2004, 2005). The wave transports the space charge, and the associated electric fields set the field lines in rotation. Near the star, the cancellation of waves from two hemispheres causes a growing region with zero poloidal current. Farther from the star, the field lines get stretched, and a pulse propagates along the equator. In an *ideal* FFE, the field lines cannot break or reconnect, and the solution tries to stretch the field lines so that they close at infinity. In the process, the magnetic field strength on the equator drops and becomes smaller than the electric field, necessitating nonideal physics, and eventually hits zero. This happens around $0.6R_{\rm LC}$ after $\frac{1}{4}$ of stellar rotation. Beyond this point, the solution forms a current sheet, supported by our resistivity prescription that keeps $E \leq B$. After one rotational period, the magnetosphere forms the closed-open configuration, but with the Y-point at $0.6R_{\rm LC}$. This is not the CKF solution, and in fact the magnetosphere proceeds to expand the closed zone to $R_{\rm LC}$ by the *reconnection* of the recently opened field lines. The rate of this expansion depends on the resistivity prescription and is 20 rotation periods of the star for our lowest resistivity model. It can be accelerated to five rotation periods if we allow for E to exceed B in the current sheet by limiting the resistive current.

Once the closed zone reaches $R_{\rm LC}$, the solution becomes quasi-steady (Fig. 1*a*), and similar to CKF and McKinney (2006b). The field lines that reconnect outside $R_{\rm LC}$ (Fig. 1*b*) break through the emission of small plasmoids every five periods, depending on resistivity, and the Y-point oscillates around $(0.95 \pm 0.05)R_{\rm LC}$. There is a sharp rise in the magnitude of the magnetic field on the equator at the Y-point (Gruzinov 2005a), followed by a jump by a factor of 10^2 to nearly 0 in the current sheet (Fig. 1*c*; the corresponding jump is a factor of 10 in Fig. 3 of McKinney 2006b, suggesting larger diffusivity). The evolution toward the light cylinder after the initial formation of the current sheet takes much longer than the light-cylinder crossing time, and the magnetosphere goes through a sequence of quasi-static equilibria similar to the states in Timokhin (2006; see our Fig. 1*c*). This evolution will be missed if code diffusivity is increased. The spin-down energy loss decreases as the closed zone expands and saturates at $(1 \pm 0.05)\mu^2\Omega_*^4/c^3$, consistent with previous work.

4. OBLIQUE PULSAR MAGNETOSPHERE

Real pulsars must have a nonzero angle between the magnetic and rotational axes in order to pulse, and so the structure of the oblique pulsar magnetosphere is most interesting. Due to the lack of symmetry, this problem has to be solved fully in three dimensions. Encouraged by the convergence of our method to known solutions for the aligned rotator, we extended our code to three dimensions. Below we discuss the gross properties of the field structure and energy loss.

We use a Cartesian grid that avoids coordinate singularities but presents difficulties in imposing boundary conditions on a central sphere. To avoid stair stepping, we force the fields to known values inside the star with a smoothing kernel (similar to Komissarov 2006). Inside the sphere, the field is set to

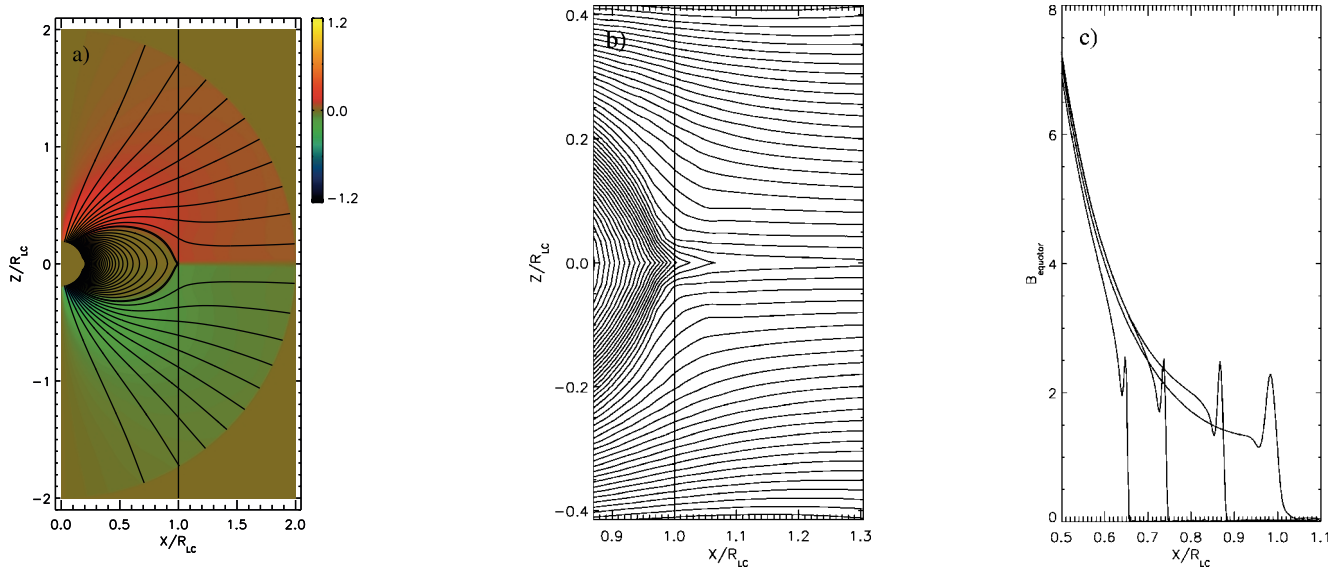

FIG. 1.—Aligned dipole magnetosphere. (*a*) Poloidal field lines of steady state solution. The thick line is the field line that touches the light cylinder, which is marked by a vertical line. Color represents the toroidal magnetic field, normalized by $\mu/(R_{\rm LC}R_*)^{3/2}$. (*b*) Zoom-in near the *Y*-point. The closed lines beyond the light cylinder are periodically thrown open by plasmoid emission. (*c*) Magnitude of the magnetic field on the equator (in units of $\mu/R_{\rm LC}^3$) during the evolution of the closed zone to the light cylinder.

$\boldsymbol{B}_0(t) = [3\hat{\boldsymbol{r}}\boldsymbol{\mu}(t)\cdot\hat{\boldsymbol{r}} - \boldsymbol{\mu}(t)]/r^3$, where $\boldsymbol{\mu}(t) = \mu(\sin\alpha\cos\Omega_* t, \sin\alpha\sin\Omega_* t, \cos\alpha)$; here α is the magnetic inclination angle. The electric field inside the sphere is forced to corotation values. We use a grid of 700^3 cells, with $R_* = 15$ cells and $R_{\rm LC} = 40$ cells. Such an unrealistic ratio $R_*/R_{\rm LC}$ is needed to resolve both the star and the wavelength $2\pi R_{\rm LC}$ on a limited grid. Without nonreflecting boundary conditions, we evolved the magnetosphere for only two turns of the star, but this seems to be enough to establish a steady state solution in the corotating frame.

Figure 2 demonstrates the force-free magnetosphere of the $\alpha = 60°$ rotator after 1.2 turns of the star. Although the magnetospheric structure is intrinsically three-dimensional, some insight can be gained from considering the shape of the field lines in the corotating frame in the plane defined by $\boldsymbol{\mu}$ and $\boldsymbol{\Omega}_*$ vectors. In this plane, the field lines are reminiscent of the CKF solution with closed and open zones, but with a current sheet that oscillates about the rotational equator in a wedge with the opening angle 2α and wavelength $2\pi R_{\rm LC}$. The field lines in this plane become straight beyond the light cylinder, similar to the inclined split-monopole solution (Bogovalov 1999). The current sheet starts at the intersection of the light cylinder with the closed zone (Fig. 2*b*), even in the oblique case. In the perpendicular plane, the field is increasingly toroidal, reversing sign in the current sheet (Fig. 3*a*).

An important question is the electromagnetic luminosity of the inclined rotator, since currently the vacuum formula $L_{\rm vac} = 2/3\mu^2\Omega_*^4/c^3\sin^2\alpha$ is commonly used to infer the magnetic field of pulsars. Our three-dimensional solution allows

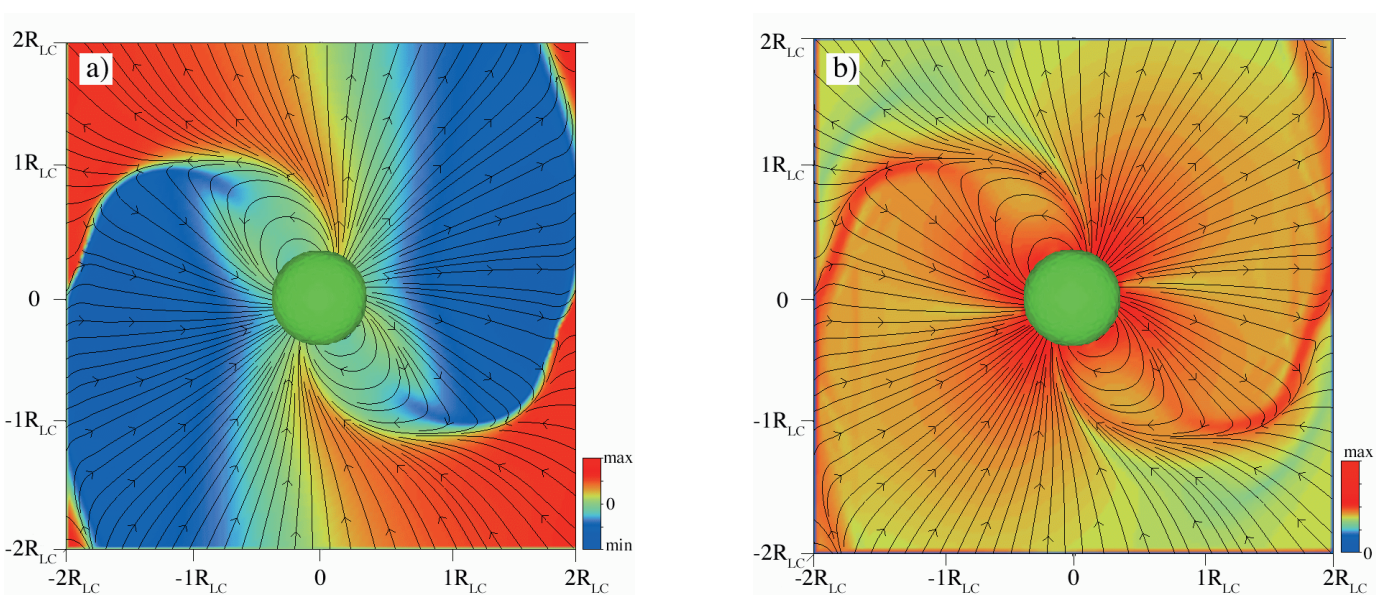

FIG. 2.—Oblique pulsar magnetosphere with magnetic inclination $\alpha = 60°$ in the corotating frame. (*a*) Magnetic field lines in the $\boldsymbol{\mu}$-$\boldsymbol{\Omega}_*$ plane. Color is the magnetic field perpendicular to the plane. (*b*) Same as (*a*), but color represents the absolute value of the total current $|\nabla\times\boldsymbol{B}|$.

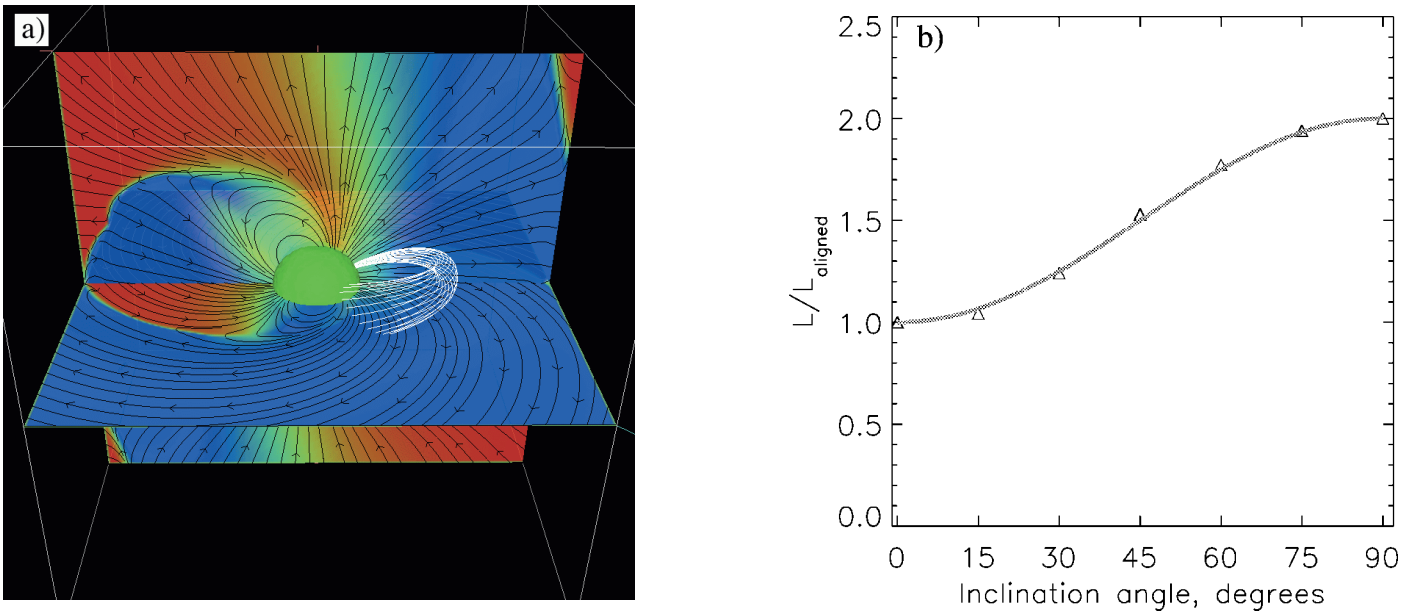

FIG. 3.—(*a*) Slices through the 60° magnetosphere. Shown are field lines in the horizontal and vertical slices; color represents the toroidal field. A sample three-dimensional flux tube is traced in white. (*b*) Spin-down luminosity in units of aligned force-free luminosity as a function of inclination. The triangles represent simulation data, and the line is a fit with eq. (3).

for a direct measurement of this quantity. We measure the Poynting flux integrated over spheres of different radius. After about one turn, the solution settles to a constant energy flux, which we plot in Figure 3*b* as a function of magnetic inclination. We find that the formula

$$L_{\rm pulsar} = k_1 \frac{\mu^2 \Omega_*^4}{c^3} (1 + k_2 \sin^2 \alpha), \qquad (3)$$

with coefficients $k_1 = 1 \pm 0.05$ and $k_2 = 1 \pm 0.1$, gives a very good fit to the oblique spin-down for all inclinations. This is roughly consistent with the estimate by Gruzinov (2005b). The inferred magnetic field at the magnetic equator of a standard neutron star is then $B_* = 2.6 \times 10^{19} (P\dot{P})^{1/2} (1 + \sin^2 \alpha)^{-1/2}$ G, which can be up to 1.7 times smaller than the estimate from the vacuum formula. The fact that oblique rotators can lose up to 2 times more power and can evolve faster in $P\dot{P}$ space than aligned rotators reinforces the prediction that near the death line there should be an excess of pulsars with smaller inclination angles (Contopoulos & Spitkovsky 2006, hereafter CS06).

We have developed a numerical method for evolving time-dependent force-free MHD equations and have applied it to solving a dynamic pulsar magnetosphere. We reproduce the closed-open configuration of CKF for the aligned rotator, and extend the solution to an oblique magnetic geometry. We find that the shape of the magnetosphere is controlled by the physics of the *Y*-point. In order for the closed zone to reach the light cylinder, reconnection must occur near the *Y*-point in the magnetosphere. If this reconnection is for some reason impeded, the magnetospheric extent may be smaller than the light cylinder. Reconnection of the open field lines must also happen when the pulsar spins down and the light cylinder recedes. If this reconnection is inefficient due to, for instance, intermittent plasmoid ejection or inertial loading, the *Y*-point may lag behind the light cylinder and lead to a braking index of less than 3 (CS06). *Y*-point dissipation also leads to time-dependent phenomena, which, when communicated to the inner magnetosphere, may leave an imprint on the radio emission (e.g., drifting subpulse phenomena). Such behavior can be modeled in our code with different prescriptions for resistivity, and further research into current sheet physics is needed. Availability of self-consistent three-dimensional magnetospheric solutions opens the way to the development of quantitative models of phenomena in the magnetospheres of pulsars, magnetars, and accretion disks.

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